FACULTY OF COMPUTING
UTM Johor Bahru

DSA Mini Project Report

SECJ2013 - Data Structures and Algorithms

Semester I 2024/2025

Section 01
Group: i_love_dsa

NAME	MATRIC NUMBER
AMELIA ADLINA BINTI AZRUL	A23CS0043
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Lecturer: **Dr. Nur Eiliyah @ Wong Yee Leng**

Date: 25 - 01 - 2025

A. Implementation of Classes and objects, Functions or File operations

<table>
<tr><th rowspan="2">Class Name</th><th rowspan="2" colspan="2">Percentage of contribution</th><th colspan="2">Location</th><th rowspan="2">Remarks</th></tr>
<tr><th>File</th><th>Line Num.</th></tr>
<tr><td rowspan="5">Queue</td><td>AMELIA ADLINA</td><td>20%</td><td rowspan="5">Queue class implementation</td><td rowspan="5">6-89</td><td rowspan="5">This class handles queue operations such as enqueue, dequeue, peek, etc.</td></tr>
<tr><td>NGU YU LING</td><td>20%</td></tr>
<tr><td>PHAVANEE KATRIYA</td><td>20%</td></tr>
<tr><td>KOO XUAN</td><td>20%</td></tr>
<tr><td>LING YU QIAN</td><td>20%</td></tr>
<tr><td rowspan="5">History</td><td>AMELIA ADLINA</td><td>20%</td><td rowspan="5">History class implementation</td><td rowspan="5">92-157</td><td rowspan="5">This class manages browser navigation history using Queue instances.</td></tr>
<tr><td>NGU YU LING</td><td>20%</td></tr>
<tr><td>PHAVANEE KATRIYA</td><td>20%</td></tr>
<tr><td>KOO XUAN</td><td>20%</td></tr>
<tr><td>LING YU QIAN</td><td>20%</td></tr>
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B. Implementation of Data Structures and Algorithms Concept (DSA)

DSA Concept	Percentage of contribution		Location		Why is this concept needed?	General idea of the implementation
			File	Line Num.		
Queue (array)	AMELIA ADLINA	20%	Queue class	6-89	The queue concept is essential in this project because it enables efficient management of browser history for forward and backward navigation. A queue operates on the First-In-First-Out (FIFO) principle, which aligns perfectly with the sequential nature of browsing history.	The implementation involves two queues: backQueue and forwardQueue. 1. Visiting a Website: When a new website is visited, the current website is enqueued into the backQueue. The forwardQueue is cleared to maintain the correct

	NGU YU LING	20%			When visiting websites, the back history grows as previous websites are stored. When navigating backward, the current website is transferred to the forward queue, enabling the user to revisit it later. This ensures a seamless browsing experience while maintaining the chronological order of history.	navigation sequence. 2. Navigating Back: The current website is enqueued into the forwardQueue, and the most recent website from the backQueue is dequeued and set as the current website. 3. Navigating Forward: The current website is enqueued into the backQueue, and the most recent website
	PHAVANEE KATRIYA	20%				

	KOO XUAN	20%				from the forwardQueue is dequeued and set as the current website. 4. Viewing History: Both queues and the current website are displayed to provide the complete navigation history.
	LING YU QIAN	20%				

C. Other Implementations (Optional)

Things / Code Done	Percentage of contribution		Remarks
Do…While (menu loop)	AMELIA ADLINA	20%	Do ... while loop is implemented in the main() function to loop the menu while the users have not exited the program. This way, users are able to perform various functions without restarting the program until they are ready to exit.
	NGU YU LING	20%	
	PHAVANEE KATRIYA	20%	
	KOO XUAN	20%	
	LING YU QIAN	20%	
Switch case (menu implementation)	AMELIA ADLINA	20%	Switch case is implemented in the main() function for selection control; users are able to select between five menu options that carry out different functions in the program.
	NGU YU LING	20%	
	PHAVANEE KATRIYA	20%	
	KOO XUAN	20%	
	LING YU QIAN	20%	
Output formatting	AMELIA ADLINA	20%	Output formatting includes adding system("cls"), spelling checks, adding newline characters where needed and prompting users to enter to continue to a new screen, for a pleasant GUI.
	NGU YU LING	20%	
	PHAVANEE KATRIYA	20%	
	KOO XUAN	20%	

	LING YU QIAN	20%	